

Landing Page A/B Test — Conversion and Revenue Validation

TripleTen Data Analytics Bootcamp · 28-day homepage experiment · 40,000 users

Statistical validation of an A/B experiment on the homepage of an e-commerce platform, run to decide whether version B should replace version A. Each comparison is matched to a test its assumptions justify — Levene before the t-test, a z-test for proportions, chi-square on counts for the segment questions — and every significant result is reported with its effect size, because at n = 40,000 significance is cheap and relevance is not. Tests that would strengthen the analysis but were not run are listed, not implied.

Business Question

Business Question:

Which version of the homepage — A (control) or B (variant) — should be implemented?

- Is there a significant difference in average spend per converted user between versions?
- Which version produces the higher conversion rate, and by how much?
- Does conversion depend on the traffic source a visitor arrives through?
- Does user type (new vs returning) influence conversion?
- How large is each difference, and is it large enough to act on?

Methodology

- Load and validate the data:** 40,000 rows x 9 columns checked for missing values, duplicate rows and repeated user IDs; category labels and column names standardized; group composition compared across region, device, channel and user type.
- Compare spending per converted user:** test the equality-of-variances assumption with Levene, then apply Welch's t-test, and report the 95% confidence interval and Cohen's d alongside the p-value.
- Review the extreme spending values** with the IQR rule inside each version, confirm they are genuine purchases rather than errors, and re-run the comparison without them as a sensitivity check before deciding whether to keep them.
- Compare conversion rates** between versions with a two-proportion z-test, reporting the difference in percentage points and in relative terms.
- Test the segment questions** with chi-square on contingency tables of counts — traffic source and user type — validating expected frequencies.
- Quantify every significant association** with Cramér's V, so that statistical significance is never read as practical relevance at n = 40,000.
- Visualize volume against effectiveness** for both categorical variables: absolute counts beside relative frequencies, since a large channel and an effective channel are different things.

```
# Data quality and consistency checks before any comparison
run_initial_checks(df)          # shape, dtypes, nulls, duplicates, repeated user_id

# Spending: the assumption check decides which t-test is valid
l_stat, p_value_var = levene(spending_A, spending_B)      # 6.9e-08 -> unequal variances
t_result = ttest_ind(spending_A, spending_B, equal_var=False)  # Welch

# Effect size and interval: the p-value says the difference exists, not how large it is
ci_low, ci_high = t_result.confidence_interval(confidence_level=0.95)
pooled_sd = np.sqrt((spending_A.var(ddof=1) + spending_B.var(ddof=1)) / 2)
cohens_d = (spending_B.mean() - spending_A.mean()) / pooled_sd

# Outliers: IQR rule inside each version, then a sensitivity re-run of the same test
q1, q3 = series.quantile([0.25, 0.75]); iqr = q3 - q1
low, high = q1 - 1.5 * iqr, q3 + 1.5 * iqr
t_trimmed = ttest_ind(trimmed_A, trimmed_B, equal_var=False)  # +7.61 vs +7.66 -> kept

# Conversion rate: two-proportion z-test
stat, pval = proportions_ztest(counts, nobs)

# Segments: chi-square on COUNTS (never on a percentage table)
contingency_table = pd.crosstab(df["traffic_source"], df["converted"])
chi2_stat, p_val, dof, expected = chi2_contingency(contingency_table)

# Effect size for the chi-square, on a 0-1 scale that does not grow with n
cramers_v = np.sqrt(chi2_stat / (len(df) * (min(contingency_table.shape) - 1)))
```

Python / pandas / scipy / statsmodels sample (data checks, test selection and effect sizes): representative, not complete — the final notebook combines this with the exploration, contingency tables and visualization steps.

Key Finding (Context → Finding → Implication)

Context

A 28-day homepage experiment exposed 40,000 users, split 19,982 / 20,018.

No missing values, no duplicate rows and no repeated users; the two groups are built from comparable audiences on region, device, channel and user type.

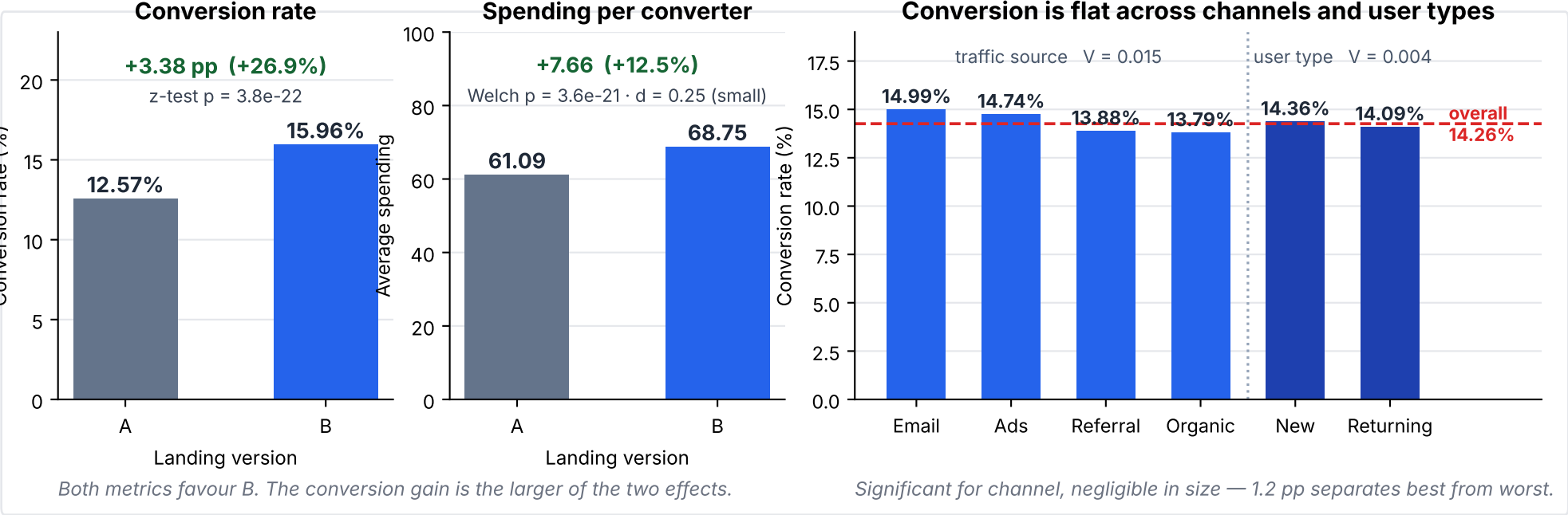
Finding

B converts 15.96% vs 12.57% — +3.38 pp, +26.9%, z-test p = 3.8e-22. Its converters also spend more: 68.75 vs 61.09, +12.5%, 95% CI [+6.08, +9.24], but Cohen's d = 0.25 marks that second effect as small. Neither segment variable matters.

Implication

Roll out B to every visitor. At the test's own traffic that is about 680 extra buyers per 20,000 visits, each spending more on average. One page for everyone — no split by channel or audience. Keep 5–10% on A for a month before calling the gain won.

Visualization



The Decision, Step by Step

1. Is the experiment trustworthy?
40,000 users split 19,982 / 20,018 · no missing values, no duplicate rows, no repeated user IDs · region, device, channel and user type distributed within 0.8 pp between the two groups

YES

2. Does B convert more visitors than A?
12.57% → 15.96% — +3.38 pp, +26.9% relative · two-proportion z-test, p = 3.8e-22 · about 680 extra converters per 20,000 visitors

YES

3. Do the visitors it converts also spend more?
61.09 → 68.75 — +7.66, +12.5%, 95% CI [+6.08, +9.24] · Welch's t-test chosen because Levene rejected equal variances (p = 6.9e-08) · Cohen's d = 0.25: the two spending distributions overlap substantially

YES, BUT SMALL

4. Does the answer change by traffic channel or user type?
Channel reaches significance ($\chi^2 = 8.66$, p = 0.034) and is still negligible: Cramér's V = 0.015, 1.2 pp separating the best channel from the worst · user type shows nothing at all (p = 0.474, V = 0.004)

NO

► IMPLEMENT LANDING PAGE B
About 680 extra buyers per 20,000 visits, each spending a little more — one page for every channel and every audience.